Ashwin V **George**

shwingeorge1@gmail.com | gregwisenoah.github.io | gregwisenoah

Education

Master of Science in Astrophysics — *Potsdam, Germany*

UNIVERSITY OF POTSDAM — *04/2023 - 03/2027 (Expected)*

- Master's thesis: Carving Cavities in the Cosmos: Simulations of AGN jets in Galaxy Groups and Clusters
- Current average: **1.5** (German scale: 1.0 = best)

Bachelor of Science in Physics — *Tamil Nadu, India*

MADRAS CHRISTIAN COLLEGE — *06/2019 - 06/2022*

- Graduated with First Class honors (GPA: **7.588**/10)

AISSCE All India Senior School Certificate Examination — *Kerala, India*

MAR DIONYSIUS SCHOOL — *03/2018*

- Average: **88.6%** (Physics: **95%**, Maths: **95%**)

Research Experience

Master's Thesis Student — *Potsdam, Germany*

LEIBNIZ INSTITUTE FOR ASTROPHYSICS — *05/2025 - 10/2026 (Submitted)*

- Conducted master's thesis research on AGN jet feedback in galaxy groups and clusters, supervised by Dr. Rainer Weinberger.
- Designed a suite of hydrodynamic simulations of galaxy groups and clusters using `AREPO`. Varied jet power, jet density, gas fraction, injection mode and validated initial conditions against IllustrisTNG.
- Showed jet-ICM coupling is sensitive to their density contrast; lighter jets entrain ICM more while heavy jets shoot through; at a fixed energy, lower power jets propagate further as higher power jets lose more energy from driving stronger shock fronts; self-regulated jets reach the virial radius and deplete gas from the halo core.
- Code: `github.com/GregwiseNoah/MSc_Thesis`

Intern — *Garching, Germany*

MAX PLANCK INSTITUTE FOR ASTROPHYSICS — *01/2025 - 05/2025*

- Set up and executed `Athena++` based hydrodynamical simulations of cold gas clouds in the intracluster medium.
- Investigated survival criteria of cold gas clouds in a turbulent, multiphase medium, under the supervision of Dr. Max Gronke and Hitesh Das; results published in Research Notes of the AAS.
- Showed that it is harder for these clouds to survive in the ICM, with survival only when $\mathrm{t}_{\mathrm{cool,mix}} / \mathrm{t}_{\mathrm{cc}} \leq 0.01$.

Publications

Survival of higher overdensity cold gas in a turbulent, multiphase medium — *Research Notes of the AAS*

Ashwin Vergis George, Hitesh Kishore Das, Max Gronke — *2025*

DOI: 10.3847/2515-5172/ae1e79, arXiv:2507.20962v2

Scholarships

DAAD Degree Completion Scholarship — *Potsdam, Germany*

German Academic Exchange Service (DAAD) – University of Potsdam — *10/2026 - 02/2027*

- Selected by a university committee based on academic performance, extracurricular engagement, and skills acquired beyond the field of study.
- Funded by the German Federal Foreign Office.

Additional Research Experience

Student Research Assistant — *Potsdam, Germany*

DEUTSCHES GEOFORSCHUNGSZENTRUM — *12/2024 - Present*

- Developing and maintaining Python pipelines to process auroral imagery and magnetometer data, working with Dr. Jürgen Matzka.
- Providing remote monitoring and technical support for various instruments at Neumayer Station III, Antarctica.

Student Research Assistant *Berlin, Germany*

HELMHOLTZ-ZENTRUM BERLIN FÜR MATERIALIEN UND ENERGIE *08/2024 - 12/2025*

- Developed micromagnetic simulations to model spin-wave propagation and skyrmion dynamics in magnetic materials with Dr. Steffen Wittrock.
- Ran GPU-accelerated simulations using BORIS Computational Spintronics.
- Designed a GUI for the real-time monitoring of low-frequency radio wave signals in superconducting cavities, visualizing forward and transmission signals.

Beamtime User at FERMI Free Electron Laser *Trieste, Italy*

ELETTRA-SINCROTRONE TRIESTE *10/2024*

- Part of an experiment to study magnons and their propagation on different materials. Contributed to data collection, analysis and experiment control of the FEL.

Conferences

Thinkshop 2025 *Potsdam, Germany*

Leibniz Institute for Astrophysics *07/2025*

- Week-long international conference on feedback in galaxy formation.
- Served on the Local Organizing Committee (LOC).

Schools

WE-Heraeus and NARIT Cosmology School 2025 – Galaxies and Beyond *Chiang Mai, Thailand*

National Astronomical Research Institute of Thailand *10/2025*

- Selected participant for a week-long international training program covering cosmology, galaxy formation, cosmological simulations, and the ΛCDM model. Funding awarded by the Wilhelm and Else Heraeus Foundation and NARIT to support travel, accommodation, and participation.
- Presentation on "The Impact of AGN Jets on their Host Galaxies."

Jürgen Ehlers Spring School on Gravitational Physics *Potsdam, Germany*

Max Planck Institute for Gravitational Physics *02/2025*

- Selected participant for spring school for introduction to gravitational physics, general relativity, black holes and gravitational-wave astrophysics.

ESCAPE Summer School *Remote*

Laboratoire d'Annecy De Physique Des Particules *06/2021*

- An online summer school for training in data science and practices relevant for astrophysics and particle physics.

Code/Astro *Remote*

Heising-Simons Foundation *06/2021*

- An online workshop conducted by astronomers from Caltech to teach fundamental software skills and best practices for building open source packages for astronomy applications.

ISSAA 2021 *Remote*

Inter-University Centre for Astronomy and Astrophysics *05/2021*

- The Introductory Summer School in Astronomy and Astrophysics is a summer school intended as a month-long introduction to the various sub-fields of astronomy and astrophysics.

Skills

Programming	Python (NumPy, SciPy, Matplotlib), C/C++, MATLAB, Bash
Simulation codes	Arepo, Athena++, BORIS Computational Spintronics
Tools	SLURM job management on HPC clusters, Git, LaTeX, Linux
Online courses	Machine learning and data analysis (Coursera, Kaggle, freeCodeCamp)

References

Dr. Rainer Weinberger
Junior Research Group Leader
Leibniz Institute for Astrophysics
Potsdam, Germany
rweinberger@aip.de

Dr. Max Gronke
ERC Research Group Leader
Astronomisches Rechen-Institut
Heidelberg, Germany
max.gronke@uni-heidelberg.de